Debashis Puhan, Ph.D, AFHEA, MInstP

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Research Profile

Tribologist and lubrication scientist with a strong focus on sustainable engineering for electrified automotive drivetrains. Currently leading an EPSRC-funded programme at the University of Sheffield on aqueous lubrication for electrified powertrains. Strong publication record (23 papers, h-index 15, 600+ citations), over 1,200 hours of university teaching, and a sustained track record of industry collaboration with BP-Castrol, Syensqo, Shell, Volvo Syensqo and Italmatch Chemicals.

Qualifications

PhD in Mechanical Engineering, Imperial College London, UK	2014–2018
M. Tech in Tribology & Maintenance Engg, Indian Institute of Technology Delhi (GPA of 9.27/10)	2011–2013
B. Tech in Mechanical Engineering, BPUT (GPA of 8.87/10)	2007–2011

Research Experience

Research Associate, University of Sheffield — Aug 2025-present

- Leading an EPSRC-funded research programme on aqueous lubrication for electric vehicle drivetrain applications, targeting motor bearings and gears.
- Investigating novel sustainable lubricants to address the unique tribological challenges of EV powertrains, including electrical current passage through contacts.
- Guest lecturer and Module Instructor for Advanced Thermodynamic Cycles (MEC-311).
- Technical Consultancy through Leonardo Testing Services. Successfully completed projects for Italmatch Chemicals and Far-UK Consultants.

Sept 2024-Mar 2025

Research & Development Engineer, Ingram Tribology

- Provided expert consulting to global clients in the lubricant, and automotive industries, delivering data-driven solutions to enhance products performance.
- Conducted standardized and custom lubricant testing, including electrified contacts in MTM and MPR (rolling contact fatigue) rigs, to support R&D of next-generation lubricants and validate durability under extreme conditions.
- Developed innovative test methodologies for rapid screening of chocolates using chocolate-artificial saliva in a pin on disc biotribometer.
- Designed the ‘Biotribology for Industry’ course for educating professionals tribological principles on biomaterials, and biosurfaces.
- Improvement of Health and Safety practices of the lab.

Research Associate, Imperial College London, UK — Apr 2023–Jun 2024

- Investigated lubricant additives (MoDTC, and ZDDP) competitive adsorption and its effect on friction and tribofilms morphology and composition to gain unique insights for engine oil formulation.
- Led an investigation of frictional behaviour of plant-based foods in collaboration with Motif FoodWorks.
- Studied the mechanical process of protein interactions during oral processes at a PDMS-PDMS interface lubricated by ex-vivo human saliva using Tribometer, Infrared spectroscopy and Electrical Impedance spectroscopy.
- Developed graphene-reinforced polyurethane (TPU) nanocomposites for water-lubricated marine stern tube bearings, investing their friction and wear performance.
- Co-supervised three Master and two PhD students for their chosen project topics related to development of alcohol based lubricant additives, water meniscus bearings, tribo-electric nano generator (TENGs), developing apparatus for studying oil aeration and foaming (Funded by Shell, UK).

Senior Research Associate, University of Cambridge, UK Oct 2022–Mar 2023

- Designed experiments and equipment for industry funded (BP Castrol) Lubricating Base Oil Project to quantify changes in the molecular structure of lubricating base oil during production.
- Designed and built a bench-top hydrocracker pressurized reactor, improving understanding of raffinate to base stock refinement process.
- Developed Thermal Diffusion Column for molecular separation in mineral oil, boosting structure-property analysis of base stocks from 5 suppliers.
- Managed a £242,000 project budget over a 3-year period, securing quotes and ensuring all of the above program, 3rd party analysis, and other costs were covered.
- Optimized process engineering for effective testing and streamlining operations, focussing on the quality, safety, and reproducibility.

Research Associate, University of Cambridge, UK May 2019- Oct 2022

- Improving the performance of existing engine oil lubricant brand and generating data for Marketing and Claims teams of BP Castrol.
- Investigated lubricant additives (Synovene, Oleamide, and ZDDP) effectiveness at the molecular level, leading to innovative next-gen lubricant with robust supply chain that benefits BP-Castrol brands and business worldwide.
- Developed methods to elucidate additive-additive interactions at the molecular level at the oil/metal interface, using HFRR and advanced techniques like Sum Frequency Generation spectroscopy and Nano Infrared microscopy.
- Engaged in confidential work for the development of next-gen lubricants and additives teaming with cross-disciplinary teams from BP-Castrol.
- Worked with Business Development and Marketing and Claims teams of BP Castrol to achieve project advancement and timely delivery of work.
- Led collaborative projects on (1) the lubrication mechanism of soft PVA hydrogel, (2) nanoscale mixing of polymer blends- PEEK and PBI + carbon fiber composites for high temperature applications yielding 2 publications.
- Co-managed two chemistry laboratories and served as the post-doctoral representative in the departmental safety committee.

Research Assistant/Associate, Imperial College London, UK May 2017–Apr 2019

- Developed high-performance composite for automotive and marine stern tube bearings.
- Studied the impact of in-situ plasma on transfer film properties in PEEK/Steel rubbing contacts, demonstrating the interplay between temperature rise, charging and plasma generation and their dependence on wear processes resulting in two publications.
- **Supervised and completed multidisciplinary collaborative research projects** leading to four journal publications, covering topics such as PEEK-PBI polymer blends, short carbon fiber-reinforced composites, acoustic emission characterization of friction and wear in dry sliding of steel, materials for battery application, wear-resistant epoxy composites for aero-application, and the development of bi-Gaussian surfaces and 3-D nanostructured materials for repelling water.

Teaching and Supervision Experience

- Engineering Subject Tutor (**550+ hours**) with three colleges of University of Cambridge - Homerton Wolfson, Robinson and Churchill College Oct 2019–Present
  - Small group teaching of Part I (1st and 2nd year) students on Structures (Stress Analysis), Mechanics, Thermofluid Mechanics and Mechanical Vibrations, class size 2-4 students.
  - Conducted undergraduate admissions interviews (2019-2021) for prospective engineering students.
  - Provided career related advice and pastoral care to students.
- Freelance Engineering Tutor (**240+ hours**), Immerse Education, UK Jul 2022–Present
  - Only in summer, I designed and delivered a two-week (40 hours)1st year Engineering UG level course for 16-18-year-olds for summer school. Student to Tutor ratio 10:1.

- Graduate Teaching Assistant (**400+ hours**), Imperial College Oct 2014–Apr 2019
  - Courses tutored: Mathematics, Materials, Mechanics, Stress Analysis, and Tribology, class size 15-20 students.
  - Marked exam papers, graded lab reports, and provided feedback.
  - Established and maintained standard procedures for smooth running of ME1 and ME2 UG laboratory work in CAD-Solid Works and Materials.
  - Co-supervised three visiting scholars, ME4/MEng research projects and ME3 DMT projects.

Funding

- Peter Jost Travel Fund Award (March 2026), £1500.
- Further funding from BP-Castrol for the ongoing research project (November 2021), £256,740 as Co-I.
- Peter Jost Travel Fund Award (March 2022), £1500.
- Dame Julia Higgins Engineering Postdoc Collaborative Research Fund (July 2018), £3,000 as Co-I.
- IOP Travel Grants (March 2017), £450.
- IOP Early Career Researchers Fund (March 2018), £300.

Awards

- Received Three Best Poster awards: 2nd Prize in Poster, STLE Tribology Frontiers Conference, Chicago, USA. Oct 2018. $200; Peter J. Blau Best Poster Award, 21st Wear of Materials, California, USA. March 2017, $250; Best Poster Award in IET Challenges in Tribology event held in Birmingham, March 2017.
- EPSRC Research Bursary for PhD (November 2013–November 2017).
- MHRD Scholarship for M. Tech program (September 2011–May 2013).

Evidence of Esteem

- Tribology Group committee Secretary at Institute of Physics, UK. Active role in organising Seminars and Workshops in UK, (2022-2026). MInstP since 2017.
- Competitively selected participant, Imperial–Tsinghua Collaborative Research Skills Development Summer School, Tsinghua University, Beijing, China (2014)
- Regularly peer-review manuscripts for international journals.
- Bye-Fellow at Wolfson College and Homerton College, Cambridge, UK.
- Post-doc representative for Department Chemistry Health & Safety Committee (2020-2023).
- Outreach services for Cambridge Science festivals, and Chemistry Open days in 2019.
- Great Exhibition Road Festival-2023 – Inspiring engineering stories for five-year-olds and Tribology group fun and demonstration activity.

Publications

Published 23 scientific research articles in international journals periodically. 600+ citations, h-index: 15; i-index: 17. Wrote 1 Book chapter and delivered talks in 10+ international conferences.

23. Jiang, S.; Puhan, D.; Kuriyagawa, K., Martin, J. M., Yamana, T., Zhang, L., Yuan, Y., Adachi, K.; Murashima, M., Tribological characteristics of phosphonium-based ionic liquids: the role of adsorption layer and tribofilms, Wear, Volumes 584–585, 206382, 2026.

22. Jiang, S.; Puhan, D.; Kuriyagawa, K., Martin, J. M., Yamana, T., Zhang, L., Yuan, Y., Adachi, K.; Murashima, M., Interactions of phosphonium-based ionic liquids and ZDDP: both synergies and antagonism (Accepted Friction)

21. Geng Z., Puhan D, Yu M., Zhang.J, Reddyhoff T., Acoustic emission monitoring of lubricant additive behaviour in sliding contacts, Friction, 2025.

20. Jiang S., Puhan D., Huang J., Yang Z., Zhang L., Yuan T., Bai X., Yuan C.; Tribological Properties of Graphene-Reinforced Polyurethane Bearing Material, Tribology International, Volume 193, 109481, 2024.

19. Jiang, S.; Wong, J.; Puhan, D.; Yuan, T.; Bai, X.; Yuan, C.; Tribological evaluation of thermoplastic polyurethane-based bearing materials under water lubrication: effect of load, sliding speed and temperature, Friction, 2023.

18. Puhan, D.; Casford, M.; Davies, P., Evaluation of Structural and Compositional Changes of a Model Mono-aromatic Hydrocarbon in a Benchtop Hydrocracker Using GC, FTIR and NMR spectroscopy, ACS Omega, 8, 39, 35988–36000, 2023.

17. Puhan D., Jiang S., Wong J., Effect of Carbon Fiber Inclusions on Polymeric Transfer Film Formation on Steel. Composites Science and Technology, 217, 109084, 2022.

16. Fellows, A.P., Puhan, D.; Wong, J.S.S., Casford, M.T.L., Davies, P.B. Probing the Nanoscale Heterogeneous Mixing in a High-Performance Polymer Blend. Polymers, 14, 192, 2022.
15. Hu S., Cao, X. Reddyhoff, T., Puhan, D., Vladescu, S.-C., Shi, X., Peng, Z., DeMello, A.-J., Dini, D., Liquid repellency enhancement through flexible microstructures. Science Advances 6, 32, 2020.
14. Hu, S., Reddyhoff, T., Puhan, D., Vladescu, S.-C., Shi, X., Dini, D., & Peng, Z., Droplet manipulation of hierarchical steel surfaces using femtosecond laser fabrication. Applied Surface Science, 521, 146474, 2020.
13. Liu, X., Ouyang, M., Orzech, M.W., Niu, Y., Tang, W., Chen, J., Marlow, M.N., Puhan, D., et al., In-situ fabrication of carbon-metal fabrics as freestanding electrodes for high-performance flexible energy storage devices. Energy Storage Materials, 30, 329-336, 2020.
12. Fellows, A. P., Puhan D., Casford M. T. L., and Davies P. B.,Understanding the Lubrication Mechanism of Polyvinyl Alcohol Hydrogels using Infrared Nanospectroscopy, Journal of Physical Chemistry C, 124, 33, 18091–18101, 2020.
11. Casford, M., Puhan, D., Davies, P., Gareth, S., Smith, T., Thermal Behaviour of Synovene and Oleamide in Oil Adsorbed on Steel, Tribology Letters 68 (2), 2020.
10. Wen, J., Reddyhoff, T., Hu, S, Puhan, D., Dini, D., Exploiting air cushion effects to optimise a superhydrophobic/hydrophilic patterned liquid ring sealed air bearing, Tribology International 144, 106129, 2020.
9. Puhan, D., Nevshupa R., Wong J. and Reddyhoff, T., Transient aspects of plasma luminescence induced by triboelectrification of polymers, Tribology International 130, 366-377, 2019.
8. Geng, Z., Puhan, D., Reddyhoff, T., Using acoustic emission to characterize friction and wear in dry sliding steel contacts, Tribology International 134, 394-407, 2019.
7. Puhan D., Wong J., Properties of Polyetheretherketone (PEEK) transferred materials in a PEEK-steel contact, Tribology International 135, 189-199, 2019.
6. Hu S., Cao, X. Reddyhoff, T, Puhan, D., Huang, W., Shi, X., Peng, Z., Dini, D. 3D-Printed Surfaces Inspired by Bi-Gaussian Stratified Plateaus. ACS Applied Materials and Interfaces 11 (22), 20528-20534, 2019.
5. Hu, S., Vladescu, S. C., Puhan, D., Huang, W., Shi, X., & Peng, Z. Bi-Gaussian stratified theory to understand wettability on rough topographies. Surface and Coatings Technology, 367, 271-277, 2019.
4. Hu, S., Reddyhoff, T., Puhan, D., Vladescu, S. C., Huang, W., Shi, X., Dini, D., Peng, Z., Bi-Gaussian Stratified Wetting Model on Rough Surfaces. Langmuir. 35, 5967–5974, 2019.
3. Hu, S., Cao, X., Reddyhoff, T., Puhan, D., Vladescu, S. C, Wang, Q., Shi, X. Peng, Z., deMello, A.J. Dini, D. Self-Compensating Liquid Repellent Surfaces with Stratified Morphology. ACS Applied Materials and Interfaces, 2019.
2. Puhan, D., Bijwe, J., Parida, T., and Trivedi, P., Investigations on Performance Properties of Nano-Micro Composites Based on Polyetherketone, Short Carbon Fibers and Hexa-Boron Nitride. Science of Advanced Materials 7, no. 5, 1002-1011, 2015.
1. Bijwe, J., Kadiyala, A. K., Kumar, K., Puhan, D., Parida, T., and Trivedi, P., Development of high-performance poly (ether-ketone) composites based on novel processing technique. Materials & Design 73, 50-59, 2015.

Book Chapter

1. Debashis Puhan *(November 27th 2020). Lubricant and Lubricant Additives, Tribology in Materials and Manufacturing - Wear, Friction and Lubrication, Amar Patnaik, Tej Singh and Vikas Kukshal, IntechOpen, DOI: 10.5772/intechopen.93830*

Conference Proceedings

11. Puhan D., Vladescu, S.C., Reddyhoff, T., Plant-Based Protein and Oral Lubrication: Investigating the Effects of Pea Protein Isolates on Salivary Films by Analysing the Frictional Response of a PDMS Oral Mimic, TriboUK, Southampton, UK, 2024.
10. Puhan D., Fellows, A. P., Casford M. T. L., and Davies P. B., Lubrication Mechanism of Poly(vinyl alcohol) Hydrogels probed by Infrared Nanospectroscopy, STLE Annual Meeting & Exhibition, Walt Disney World Swan and Dolphin Resort, Orlando, Florida 2022.
9. Wong, J., Puhan, D., High Temperature Tribology of PEEK-blends and PEEK-composites in Unlubricated Conditions, International Tribology Conference, Sendai 2019, Sendai International Center, Sendai, Japan, 17-21 September 2019.
8. Reddyhoff, T., Geng, Z., Puhan, D.,Characterising Tribological Behaviour in Dry Sliding Steel Contacts Using Acoustic Emission, International Tribology Conference, Sendai 2019, Sendai International Center, Sendai, Japan, 17-21 September 2019.
7. Puhan D., Wong J., Micro mechanisms of Transfer Film Formation and Sliding Wear of PEEK, STLE Tribology Frontiers Conference 2018, Chicago, USA, 28-31 October 2018.
6. Puhan, D. and Reddyhoff, T., Transient characteristics of triboplasma generation, Proceedings of the 43rd Leeds-Lyon Symposium on Tribology, Leeds, United Kingdom, 6-9 September 2016.
5. Puhan, D. and Reddyhoff T., Triboplasma generation at polymer contacts, Proceedings of the 42th Leeds-Lyon Symposium on Tribology, Lyon, France 7-9 September 2015.

4. Puhan, D. and Reddyhoff T., An Experimental Study into the Effects of Triboemission, International Tribology Conference, Tokyo 2015, Tokyo University of Science, Tokyo, Japan, 16-20 September 2015.
3. Reddyhoff, T. and Puhan, D., Transient Triboplasma Measurements, International Tribology Conference, Tokyo 2015, Tokyo University of Science, Tokyo, Japan, 16-20 September 2015.
2. Bijwe, J., Puhan, D., Kadiyala A. K., Parida, T., and Trivedi, P., Dependence of Performance properties of Polyaryletherketone (PAEK) composites on processing parameters. 1st International Conference on Mechanics of Composites (MECHCOMP2014), Long Island, NY State, USA, 8-12 June 2014.
1. Puhan, D., Bijwe, J., Kadiyala A. K., Parida, T., and Trivedi, P., Influence of process of dispersion of solid lubricant in composite on performance properties of Polyaryletherketone (PAEK) composites. ASIATRIB- 2013.